

SLA Compliance Analysis for LATIS Corrective Maintenance Tickets

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Abstract

This report documents the development of an Alteryx workflow to analyze Service Level Agreement (SLA) compliance for corrective maintenance tickets within the LATIS sub-project. It details data preparation, filtering, transformations, SLA logic, aggregation, and output structuring, addressing challenges and presenting deliverables.

1 Introduction

This report outlines the steps followed to design and implement an Alteryx workflow aimed at analyzing Service Level Agreement (SLA) compliance for corrective maintenance tickets within the LATIS sub-project. The goal was to clean the data, apply business logic, calculate SLA Key Performance Indicators (KPIs), and produce structured summary outputs. The project utilizes two provided Excel files: *Full PCM History* (containing all fault tickets) and *Excluded Work Orders* (listing orders to exclude), with specific requirements outlined in the document *SLA_PoD_Corrective_Maintenance_LATIS_Task.pdf*. This report details each step, the technical configurations, challenges encountered, and the final deliverables.

2 Data Preparation

2.1 Objective

Import the Excel files and exclude the specified work orders.

2.2 Steps Performed

- **File Importation:** Used an *Input Data* tool to import *Full PCM History.xlsx* (containing all fault tickets) and *Excluded Work Orders.xlsx* (list of orders to exclude).
- **Exclusion of Orders:** Added a *Join* tool to compare the *Order ID* columns between the two files, configured in "Left Exclude" mode to retain only tickets not listed in *Excluded Work Orders*.

2.3 Result

A cleaned dataset containing only valid tickets.

2.4 Challenges

No major issues were reported, but manual verification was recommended to ensure all excluded *Order ID* values were correctly identified.

3 Application of Filters

3.1 Objective

Apply filters to select relevant tickets based on the specified criteria.

3.2 Steps Performed

- **Addition of a Filter Tool:** Configured with the following conditions:
 - *[Status]* DOES NOT CONTAIN "Escalated" OR "Cancel".
 - *[Sub Project]* = "LATIS".
 - *[TT Type]* IN ("EM", "Reference", "EM", "Telecom", "Reference", "Telecom").
 - *[Title]* DOES NOT CONTAIN "HC", "PT", "PM Error", "Performance Issue", "Visibility", "Dust", "Chassis", "Health".
- **Execution:** Tested with a *Browse* tool to validate the dataset reduction.

3.3 Result

Filtered dataset containing only tickets meeting the criteria.

3.4 Challenges

Duplication in *[TT Type]* ("EM", "Reference") required clarification; interpreted as a valid list with accepted redundancies.

4 Field Transformations

4.1 Objective

Convert and calculate fields to prepare the data for SLA analysis.

4.2 Steps Performed

- **Type Conversions:** Used a *Select* tool to convert text fields (e.g., *Fault Level*, *Impact*) to *V_WString* and *Created At* and *Fault Recovery Time* to *DateTime*.
- **Duration Calculation:** Added a *Formula* tool with the expression: *[Duration (seconds)] = DateTimeDiff([Fault Recovery Time], [Created At], "seconds")*.
- **NSA Impact Logic:** Added a *Formula* tool to create *[Affected?]* with the expression: *IIF(Contains([Impact], "NSA"), "NSA", "SA")*; updated *[Fault Level]* with: *IIF([Affected?] = "NSA", "Minor", [Fault Level])*.

4.3 Result

Dataset with converted fields and applied impact logic.

4.4 Challenges

Handling null values in *Fault Recovery Time* required additional validation.

5 SLA Threshold Logic

5.1 Objective

Determine whether tickets meet the Target and Hurdle SLA thresholds.

5.2 Steps Performed

- **Creation of Target and Hurdle Fields:** Added a *Formula* tool for *[Target]* with the expression:

```
IIF([Fault Level]="Emergency" AND [Duration]>7200, "NOK",
IIF([Fault Level]="Critical" AND [Duration]>14400, "NOK",
IIF([Fault Level]="Major" AND [Duration]>28800, "NOK",
IIF([Fault Level]="Minor" AND [Duration]>43200, "NOK",
IIF([Resolved Owner]=" ", "NOK",
IIF([Fault Recovery Time]="N/A" OR [Fault Recovery Time]=" " OR IsNull([Fault
```

Added a *Formula* tool for *[Hurdle]* with similar thresholds (14400, 21600, 43200, 86400 seconds).

5.3 Result

Fields *[Target]* and *[Hurdle]* marked "NOK" for violations, "" otherwise.

5.4 Challenges

Ensuring consistency between duration values (in seconds) and the provided benchmarks (in hours).

6 Data Aggregation

6.1 Objective

Group data by severity level and compute metrics.

6.2 Steps Performed

- **Summarize Tool:** "Group By" on *[Fault Level]*, calculations for *Total PCMs* (Count on *[Order ID]*), *Within Target SLA* (Count If *[Target] = ""*), *Exceeded Target SLA* (Count If *[Target] = "NOK"*), *Within Hurdle SLA* (Count If *[Hurdle] = ""*), *Exceeded Hurdle SLA* (Count If *[Hurdle] = "NOK"*).
- **Correction for "Emergency":** Added a *Text Input* with *Emergency*, 0, 0, 0, 0, 0, used a *Union* tool before *Summarize*.

6.3 Result

Aggregated table including "Emergency".

6.4 Challenges

Initial absence of "Emergency" addressed through manual addition.

7 SLA Percentage Calculations

7.1 Objective

Calculate SLA compliance rates.

7.2 Steps Performed

- **Formula Tool:** *Achieved Target SLA %* = *IIF*(*[Total PCMs] = 0*, "-", *ToString(Round([Within Target SLA] / [Total PCMs] * 100, 2)) + "%"*) (FixedDecimal and V_WString); similar for *Achieved Hurdle SLA %*.

7.3 Result

Calculated percentages (e.g., 97.69% for Minor Target).

7.4 Challenges

Potential inconsistency with PDF data (98.15% for Hurdle Minor) requiring validation.

8 Join with SLA Benchmark Table

8.1 Objective

Add SLA benchmarks to the dataset.

8.2 Steps Performed

- **Join Tool:** Joined on $[Fault\ Level] = [Severity]$ with the benchmark table:

Severity	Target	Hurdle
Emergency	90% < 2 hours	96% < 4 hours
Critical	90% < 4 hours	95% < 6 hours
Major	90% < 8 hours	95% < 12 hours
Minor	90% < 12 hours	95% < 24 hours

Added via *Text Input* or *Join Multiple*.

- **Sort Tool:** Manual sort ("Emergency", "Critical", "Major", "Minor").

8.3 Result

Enriched data with benchmarks.

8.4 Challenges

None, but manual validation recommended.

9 Final Output Columns

9.1 Objective

Structure the output according to requirements.

9.2 Steps Performed

- **Select Tool:** Order: *Fault Level*, *Total PCMs*, *Within/Exceeded SLA (Target)*, *Achieved SLA % (Target)*, *Target Benchmark*, *Within/Exceeded SLA (Hurdle)*, *Achieved SLA % (Hurdle)*, *Hurdle Benchmark*.

10 Preparation of Deliverables

10.1 Objective

Provide the requested files.

10.2 Steps Performed

- **Workflow (.yxmd):** Saved via *File > Save As* (e.g., *Final_SLA_Workflow.yxmd*).
- **Output Dataset (.xlsx):** Exported via *Output Data* (e.g., *SLA_Output.xlsx*).
- **Screenshots:** Instructions to capture *Input*, *Summarize*, *Select*, etc.

10.3 Result

Files ready for delivery.

11 Conclusion

This project successfully developed a comprehensive Alteryx workflow to analyze SLA compliance for LATIS tickets, adhering to data preparation, filtering, transformation, SLA logic, aggregation, and output structuring steps. Key challenges (absence of "Emergency", PDF format) were addressed with alternative solutions. The deliverables (workflow, dataset, screenshots) are prepared, but the PDF export requires a final decision on format (improved table or narrative report). Future improvements could include automating PDF conversion and validating data against the source PDF. This report serves as a complete documentation of the project.

12 Appendices

12.1 Files Used

- *Full PCM History.xlsx*
- *Excluded Work Orders.xlsx*

12.2 References

- *SLA_PoD_Corrective_Maintenance_LATIS_Task.pdf*